ChatGPT - Lecture Scribe 5, 9. 13, 17

L21_L22.pdf
PDF

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Lecture Scribe

CSE400 – Fundamentals of Probability in Computing

Lecture 21 and 22: Inequalities and Tail Bound

This lecture introduces tails, motivates the need for tail bounds, presents two tail-event examples, develops a running example based on a binomial random variable, and states and applies Markov's inequality.
L21_L22

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1. Outline

The lecture covers the following topics:

- Tail and Motivation of Tail Bound
- Markov's Inequality
  - Definition and Proof
  - Example
- Chebyshive's Inequality
  - Definition and Proof
  - Example
- Chernof Bound and Poison Tail
  - Definition and Proof
  - Example
- Jensen's Inequality
  - Convex Functions
  - Proof and Example
- Case Study and System Level Understanding

L21_L22

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2. Tails

2.1 Definition

The tail of a random variable X is defined as

$$P\{X > x\}.$$

This is the probability that the random variable exceeds the value x. L21_L22

2.2 Why tails matter

The lecture gives the following motivating examples:

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer L21_L22

2.3 Key idea

The lecture distinguishes between mean behavior and tail behavior as follows:

- Mean $\longrightarrow$ average behavior (easy)
- Tail $\longrightarrow$ rare extreme events (hard) L21_L22

2.4 Why tails are hard to compute

The lecture states that tails are hard to compute because they require summing many small probabilities. L21_L22

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3. Tail Example 1

3.1 Statement of the example

Suppose n jobs are distributed among n servers at random. The question is:

What is the probability that a particular server gets at least k jobs? L21_L22

3.2 Event description

The lecture presents the following points:

- Jobs are assigned randomly
- Most servers get few jobs
- Occasionally, one server gets many jobs

This is identified as a tail event, described as a rare overload. L21_L22

3.3 Distribution and tail probability

The lecture models the number of jobs at the highlighted server as

$$X \sim \text{Binomial}(n, p).$$

The required probability is

$$P\{X \geq k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}.$$

This expression is presented as the exact tail probability for the event that a particular server gets at least k jobs. L21_L22

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4. Tail Example 2

4.1 Statement of the example

Jobs arrive to a datacenter according to a Poisson process with rate λ jobs/hour. The question is:

What is the probability that at least k jobs arrive during the first hour? L21_L22

4.2 Event description

The lecture presents the following points:

- Jobs arrive randomly over time
- Most hours have moderate arrivals
- Sometimes, we see a burst of arrivals

This is identified as a tail event, described as a sudden spike in load. L21_L22

4.3 Distribution and tail probability

The lecture models the number of arrivals in one hour as

$$X \sim \text{Poisson}(\lambda).$$

The required probability is

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}.$$

This is presented as the exact tail probability for observing at least k arrivals in the first hour. L21_L22

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5. Why Tail Bounds?

5.1 Difficulty of exact computation

The lecture states that exact tail probabilities are often hard to compute. Two exact computations are shown:

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

and

$$P\{X \geq k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}.$$

These are described as complicated, long sums. L21_L22

5.2 Main idea of a tail bound

Instead of computing the exact value, the lecture proposes finding an upper bound. The reasons given are:

- We find an upper bound
- Easier to compute
- Still gives a guarantee L21_L22

This is summarized by the comparison:

- Exact: $P(X \geq k) = ?$
- Bound: $P(X \geq k) \leq$ something simple L21_L22

The key idea stated in the lecture is:

Approximate safely instead of computing exactly. L21_L22

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6. Running Example

6.1 Purpose

The lecture introduces a running example with two goals:

- Develop progressively better (tighter) tail bounds
- Test all bounds on the same example L21_L22

6.2 Distribution

The running example uses

$$X \sim \text{Binomial}\left(n, \frac{1}{2}\right).$$

The accompanying figure labels the mean as

$$\mu = \frac{n}{2}.$$

The tail region of interest is shown as

$$P(X \geq 3n/4),$$

described as getting unusually many heads. L21_L22

6.3 Experiment, distribution, mean, and question

The lecture states:

Experiment: Flip a fair coin n times.

Distribution:

$$X = \text{number of heads}$$

$$X \sim \text{Binomial}\left(n, \frac{1}{2}\right)$$

Mean:

$$E[X] = \frac{n}{2}$$

Question:

$$\text{What is } P\left(X \geq \frac{3}{4}n\right)?$$

L21_L22

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7. Markov's Inequality

7.1 Key idea

The lecture presents Markov's inequality through the following statements:

- Uses only the mean
- Large values $\Rightarrow$ rare

L21_L22

7.2 Example used for motivation

The lecture gives the example:

Average $= 10 \Rightarrow$ few values can be $100+$.

L21_L22

7.3 Statement of Markov's inequality

The inequality is stated as

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

L21_L22

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8. Proof Idea for Markov's Inequality

The lecture presents the proof idea in the following steps:

1. Consider only $X \geq a$.
2. Replace X by a.

L21_L22

Then the derivation shown is

$$\begin{aligned} E[X] &= \sum_{x=0}^{\infty} x \cdot p_X(x) \\ &\geq \sum_{x=a}^{\infty} x \cdot p_X(x) \\ &\geq \sum_{x=a}^{\infty} a \cdot p_X(x) \\ &= a \cdot P(X \geq a). \end{aligned}$$

From this displayed chain, the stated inequality follows:

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

L21_L22

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9. Markov's Inequality on the Running Example

9.1 Problem statement

For the experiment of flipping a fair coin n times, the lecture asks:

What is a tail bound on the probability of getting at least $\frac{3}{4}n$ heads? L21_L22

9.2 Application of Markov's inequality

The lecture uses:

$$X = \text{Number of Heads} \sim \text{Binomial}\left(n, \frac{1}{2}\right)$$

and

$$\mu = E[X] = \frac{n}{2}.$$

Applying Markov's inequality to the event $X \geq \frac{3}{4}n$,

$$P\left\{X \geq \frac{3}{4}n\right\} \leq \frac{\mu}{\frac{3}{4}n} = \frac{\frac{n}{2}}{\frac{3}{4}n} = \frac{2}{3}.$$

L21_L22

9.3 Evaluation of the bound

The lecture explicitly states:

Clearly a terrible bound because doesn't involve n. L21_L22

10. Markov Inequality: Real-Life Example

The lecture presents the following scenario:

Scenario: Income Distribution

- Average income $= 10{,}000$ L21_L22

No further derivation or conclusion is shown in the provided slide content excerpt. L21_L22

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11. Case Study

A later slide states:

Case Study: How to predict the worst case demands on server? L21_L22

No additional case-study development is shown in the provided slide content excerpt. L21_L22

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12. Closing

The lecture file ends with:

Thank You :) L21_L22

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CSE400 – Fundamentals of Probability in Computing

Lecture 21 and 22: Inequalities and Tail Bound

L21_L22

Outline

The lecture covers the following topics:

- Tail and Motivation of Tail Bound
- Markov's Inequality
  - Definition and Proof
  - Example
- Chebyshive's Inequality
  - Definition and Proof
  - Example
- Chernof Bound and Poison Tail
  - Definition and Proof

    - Example
- Jensen's Inequality
    - Convex Functions
    - Proof and Example
- Case Study and System Level Understanding

L21_L22

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1. Tails

Definition

The tail of a random variable X is

$$P\{X > x\}.$$

Why do we care about tails?

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

Key idea

- Mean $\rightarrow$ average behavior (easy)
- Tail $\rightarrow$ rare extreme events (hard)

Why are tails hard to compute?

- Requires summing many small probabilities

L21_L22

---

2. Tails Example-1/2

Question

Suppose you're distributing n jobs among n servers at random. What's the probability that a particular server gets $\geq k$ jobs?

Statements given in the lecture

- Jobs are assigned randomly
- Most servers get few jobs
- Occasionally, one server gets many jobs

This is a tail event (rare overload).

Distribution

$$X \sim \text{Binomial}(n, p)$$

Tail probability

$$P\{X \geq k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

The highlighted server has $\geq k$ jobs. L21_L22

---

3. Tails Example-2/2

Question

Jobs arrive to a datacenter according to a Poisson process with rate λ jobs/hour. What's the probability that $\geq k$ jobs arrive during the first hour?

Statements given in the lecture

- Jobs arrive randomly over time
- Most hours have moderate arrivals
- Sometimes, we see a burst of arrivals

This is a tail event (sudden spike in load).

Distribution

$$X \sim \text{Poisson}(\lambda)$$

Tail probability

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \cdot \frac{\lambda^i}{i!}$$

This corresponds to many arrivals within one hour $(\geq k \text{ jobs})$. L21_L22

---

4. Why Tail Bounds?

Exact tail probabilities are often hard to compute.

Exact computation

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \cdot \frac{\lambda^i}{i!}$$

$$P\{X \geq k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Complicated, long sums.

Idea: Instead of exact value

- We find an upper bound
- Easier to compute
- Still gives a guarantee

$$\text{Exact: } P(X \geq k) = ?$$

$$\text{Bound: } P(X \geq k) \leq \text{something simple}$$

Key idea

Approximate safely instead of computing exactly.
L21_L22

---

5. Running Example (Key Idea)

The lecture uses one running example so that:

- We will develop progressively better (tighter) tail bounds
- We test all bounds on the same example

Distribution used in the running example

$$X \sim \text{Binomial}(n, 1/2)$$

The mean is marked as

$$\mu = n/2.$$

The tail region shown is

$$P(X \geq 3n/4)$$

and is described as getting unusually many heads.
L21_L22

---

6. Running Example: Full Statement

Experiment

Flip a fair coin n times.

Distribution

$$X = \text{number of heads}$$

$$X \sim \text{Binomial}(n, \tfrac{1}{2})$$

Mean

$$E[X] = \frac{n}{2}$$

Question

What is

$$P\left(X \geq \frac{3}{4}n\right)?$$

L21_L22

---

7. Markov's Inequality

Key idea

- Uses only the mean
- Large values $\Rightarrow$ rare

Example statement

Avg $= 10 \Rightarrow$ few values can be $100+$

Markov Inequality

$$P(X \geq a) \leq \frac{E[X]}{a}$$

L21_L22

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8. Proof (Idea) of Markov's Inequality

The proof idea is presented in the following steps:

- Consider only $X \geq a$
- Replace X by a

Then the derivation shown is

$$E[X] = \sum_{x=0}^{\infty} x \cdot p_X(x)$$

$$\geq \sum_{x=a}^{\infty} x \cdot p_X(x)$$

$$\geq \sum_{x=a}^{\infty} a \cdot p_X(x)$$

$$= a \cdot P(X \geq a)$$

Hence the stated inequality is

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

L21_L22

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9. Markov's Inequality on Running Example

Question

Flip a fair coin n times:

What's a tail bound on the probability of getting at least $\frac{3}{4}n$ heads?

Given quantities

$$X = \text{Number Heads} \sim \text{Binomial}\left(n, \frac{1}{2}\right)$$

$$\mu = E[X] = \frac{n}{2}$$

Applying Markov's inequality

$$P\left\{X \geq \frac{3}{4}n\right\} \leq \frac{\mu}{\frac{3}{4}n} = \frac{\frac{n}{2}}{\frac{3}{4}n} = \frac{2}{3}$$

Conclusion stated in the lecture

Clearly a terrible bound because doesn't involve n.

L21_L22

---

10. Markov Inequality: Real-Life Example

Scenario

Income Distribution

- Average income $= 10{,}000$

L21_L22

---

11. Case Study

The lecture states:

12. Closing

Thank You :)
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  - Convex Functions
  - Proof and Example
- Case Study and System Level Understanding L21_L22

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1. Tails

Definition

The tail of a random variable X is

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Why do we care about tails?

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

Key idea

- Mean $\rightarrow$ average behavior (easy)
- Tail $\rightarrow$ rare extreme events (hard)

Why are tails hard to compute?

- Requires summing many small probabilities

L21_L22

---

2. Tails Example-1/2

Question

Suppose you're distributing n jobs among n servers at random. What's the probability that a particular server gets $\geq k$ jobs?

Statements presented in the lecture

- Jobs are assigned randomly
- Most servers get few jobs
- Occasionally, one server gets many jobs

This is a tail event (rare overload).

Distribution

$$X \sim \text{Binomial}(n, p)$$

Tail probability

$$P\{X \geq k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

The highlighted server has $\geq k$ jobs.

L21_L22

---

3. Tails Example-2/2

Question

Jobs arrive to a datacenter according to a Poisson process with rate λ jobs/hour. What's the probability that $\geq k$ jobs arrive during the first hour?

Statements presented in the lecture

- Jobs arrive randomly over time
- Most hours have moderate arrivals
- Sometimes, we see a burst of arrivals

This is a tail event (sudden spike in load).

Distribution

$$X \sim \text{Poisson}(\lambda)$$

Tail probability

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \cdot \frac{\lambda^i}{i!}$$

This corresponds to many arrivals within one hour $(\geq k \text{ jobs})$. L21_L22

---

4. Why Tail Bounds?

Exact tail probabilities are often hard to compute.

Exact computation

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \cdot \frac{\lambda^i}{i!}$$

$$P\{X \geq k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Complicated, long sums.

Idea: Instead of exact value

- We find an upper bound
- Easier to compute
- Still gives a guarantee

$$\text{Exact: } P(X \geq k) = ?$$

$$\text{Bound: } P(X \geq k) \leq \text{something simple}$$

Key idea

Approximate safely instead of computing exactly. L21_L22

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5. Running Example (Key Idea)

The lecture introduces one running example with the following purpose:

- We will develop progressively better (tighter) tail bounds
- We test all bounds on the same example

Distribution used in the running example

$$X \sim \text{Binomial}(n, 1/2)$$

The mean is indicated as

$$\mu = n/2.$$

The tail region is shown as

$$P(X \geq 3n/4)$$

and is labeled as getting unusually many heads.

L21_L22

---

6. Running Example: Full Statement

Experiment

Flip a fair coin n times.

Distribution

$$X = \text{number of heads}$$

$$X \sim \text{Binomial}(n, \tfrac{1}{2})$$

Mean

$$E[X] = \frac{n}{2}$$

Question

$$\text{What is } P\left(X \geq \frac{3}{4}n\right)?$$

L21_L22

---

7. Markov's Inequality

Key idea

- Uses only the mean
- Large values $\Rightarrow$ rare

Example

$$\text{Avg} = 10 \Rightarrow \text{few values can be } 100+$$

Statement

$$P(X \geq a) \leq \frac{E[X]}{a}$$

L21_L22

---

8. Proof (Idea) of Markov's Inequality

The lecture presents the proof idea in the following steps:

- Consider only $X \geq a$
- Replace X by a

Then the derivation shown is

$$\begin{aligned} E[X] &= \sum_{x=0}^{\infty} x \cdot p_X(x) \\ &\geq \sum_{x=a}^{\infty} x \cdot p_X(x) \\ &\geq \sum_{x=a}^{\infty} a \cdot p_X(x) \\ &= a \cdot P(X \geq a) \end{aligned}$$

Hence,

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

L21_L22

---

9. Markov's Inequality on Running Example

Question

Flip a fair coin n times:

What's a tail bound on the probability of getting at least $\frac{3}{4}n$ heads?

Given

$$X = \text{Number Heads} \sim \text{Binomial}\left(n, \frac{1}{2}\right)$$

$$\mu = E[X] = \frac{n}{2}$$

Applying Markov's inequality

$$P\left\{X \geq \frac{3}{4}n\right\} \leq \frac{\mu}{\frac{3}{4}n} = \frac{\frac{n}{2}}{\frac{3}{4}n} = \frac{2}{3}$$

Conclusion stated in the lecture

Clearly a terrible bound because doesn't involve n.
L21_L22

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10. Markov Inequality: Real-Life Example

Scenario

Income Distribution

- Average income $= 10{,}000$
L21_L22

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11. Case Study

The lecture states:

How to predict the worst case demands on server?
L21_L22

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12. Closing

Thank You :)
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L21_L22

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1. Tails

1.1 Definition

The tail of a random variable X is

$$P\{X > x\}.$$

1.2 Why do we care about tails?

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

1.3 Key idea

- Mean $\rightarrow$ average behavior (easy)
- Tail $\rightarrow$ rare extreme events (hard)

1.4 Why are tails hard to compute?

- Requires summing many small probabilities

L21_L22

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2. Tails Example-1/2

2.1 Question

Suppose you're distributing n jobs among n servers at random. What's the probability that a particular server gets $\geq k$ jobs?

L21_L22

2.2 Statements shown in the lecture

- Jobs are assigned randomly
- Most servers get few jobs
- Occasionally, one server gets many jobs

This is a **tail event** (rare overload). L21_L22

2.3 Distribution

$$X \sim \text{Binomial}(n, p)$$

2.4 Tail probability

$$P\{X \geq k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

The highlighted server has $\geq k$ jobs. L21_L22

---

3. Tails Example-2/2

3.1 Question

Jobs arrive to a datacenter according to a Poisson process with rate λ jobs/hour. What's the probability that $\geq k$ jobs arrive during the first hour? L21_L22

3.2 Statements shown in the lecture

- Jobs arrive randomly over time
- Most hours have *moderate* arrivals
- Sometimes, we see a **burst of arrivals**

This is a **tail event** (sudden spike in load). L21_L22

3.3 Distribution

$$X \sim \text{Poisson}(\lambda)$$

3.4 Tail probability

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \cdot \frac{\lambda^i}{i!}$$

This corresponds to many arrivals within one hour $(\geq k \text{ jobs})$. L21_L22

---

4. Why Tail Bounds?

Exact tail probabilities are often **hard to compute**. L21_L22

4.1 Exact computation

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \cdot \frac{\lambda^i}{i!}$$

$$P\{X \geq k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Complicated, long sums. L21_L22

4.2 Idea: Instead of exact value

- We find an upper bound
- Easier to compute
- Still gives a guarantee

$$\text{Exact: } P(X \geq k) = ?$$

$$\text{Bound: } P(X \geq k) \leq \text{something simple}$$

4.3 Key idea

Approximate safely instead of computing exactly. L21_L22

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5. Running Example (Key Idea)

The lecture introduces one running example so that:

- We will develop progressively better (tighter) tail bounds
- We test all bounds on the same example L21_L22

5.1 Distribution used

$$X \sim \text{Binomial}(n, 1/2)$$

The mean is indicated as

$$\mu = n/2$$

The tail region shown is

$$P(X \geq 3n/4)$$

and the tail region is identified as getting unusually many heads. L21_L22

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6. Running Example: Full Statement

6.1 Experiment

Flip a fair coin n times.

6.2 Distribution

$$X = \text{number of heads}$$

$$X \sim \text{Binomial}\left(n, \frac{1}{2}\right)$$

6.3 Mean

$$E[X] = \frac{n}{2}$$

6.4 Question

$$\text{What is } P\left(X \geq \frac{3}{4}n\right)?$$

L21_L22

---

7. Markov's Inequality

7.1 Key Idea

- Uses only the mean
- Large values $\Rightarrow$ rare

7.2 Example

$$\text{Avg} = 10 \Rightarrow \text{few values can be } 100+$$

7.3 Statement

$$P(X \geq a) \leq \frac{E[X]}{a}$$

L21_L22

---

8. Proof (Idea) of Markov's Inequality

The lecture presents the following proof idea:

- Consider only $X \geq a$
- Replace X by a

Then,

$$
\begin{aligned}
E[X] &= \sum_{x=0}^{\infty} x \cdot p_X(x) \\
&\geq \sum_{x=a}^{\infty} x \cdot p_X(x) \\
&\geq \sum_{x=a}^{\infty} a \cdot p_X(x) \\
&= a \cdot P(X \geq a)
\end{aligned}
$$

Hence,

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

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9. Markov's Inequality on Running Example

9.1 Question

Flip a fair coin n times:

What's a tail bound on the probability of getting at least $\frac{3}{4}n$ heads? L21_L22

9.2 Given

$$X = \text{Number Heads} \sim \text{Binomial}\left(n, \frac{1}{2}\right)$$

$$\mu = E[X] = \frac{n}{2}$$

9.3 Applying Markov's inequality

$$P\left\{X \geq \frac{3}{4}n\right\} \leq \frac{\mu}{\frac{3}{4}n} = \frac{\frac{n}{2}}{\frac{3}{4}n} = \frac{2}{3}$$

9.4 Conclusion stated in the lecture

Clearly a terrible bound because doesn't involve n. L21_L22

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10. Markov Inequality: Real-Life Example

Scenario

Income Distribution

- Average income $= 10{,}000$ L21_L22

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11. Case Study

The lecture states:

How to predict the worst case demands on server? L21_L22

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12. Closing

Thank You :) L21_L22

Sources